

REVIT FOR STRUCTURES

Steel and concrete structural modeling in Autodesk Revit. The full program case study, module by module, across two live online tracks.

2

TRAINING TRACKS

49

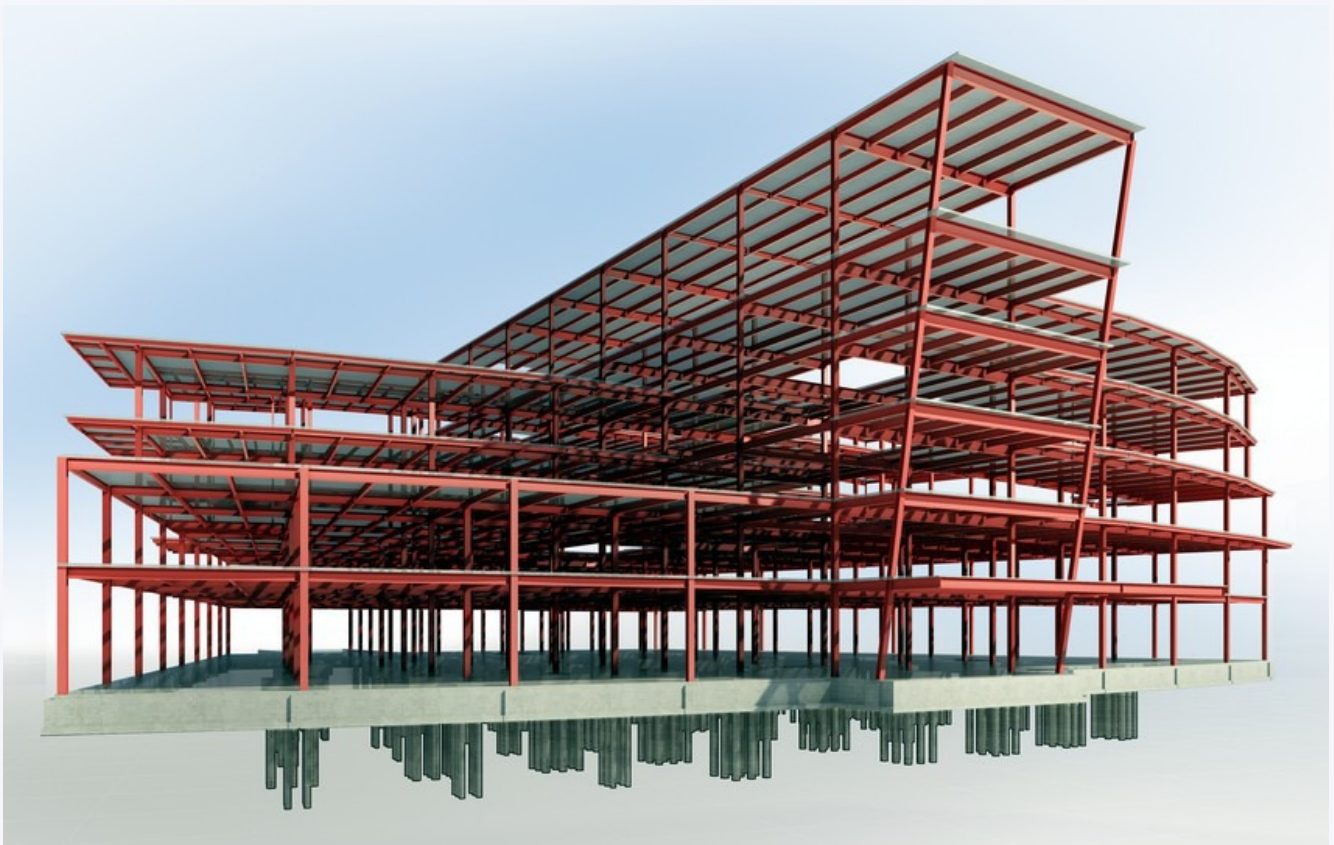
MODULES IN FULL

173

SUBTOPICS

4 wks

PER TRACK ONLINE



Background

Revit for Structures is a hands on Autodesk Revit training program for engineering students and early career professionals who want to build real, buildable structural models, not just pretty renders. The program runs as two focused tracks that can be taken separately or together: Reinforced Concrete Structures and Steel Structures.

Both tracks move from the fundamentals of the Revit interface through full structural modeling, reinforcement and connection detailing, schedules and construction documentation.

START DATE

Monday, 14 September 2026, third week of September

FORMAT

100% online, live sessions led by an instructor on Zoom

DURATION

4 weeks per track, weekday evenings

FEE

KSH 2,500 per track

WHO IT'S FOR

University engineering students and early career structural and BIM professionals

PREREQUISITE

A laptop capable of running Revit, no prior Revit experience required

The challenge

01

Who it is for

University engineering students and early career structural and BIM professionals who want to build real, buildable models, not just pretty renders.

02

Starting level

No prior Revit experience is required. The only prerequisite is a laptop capable of running Autodesk Revit.

03

Format and pace

100% online, live sessions led by an instructor on Zoom, weekday evenings, four weeks per track.

04

The gap it closes

Moving from knowing Revit exists to modeling complete structures end to end, with reinforcement, connections, schedules and sheets.

The solution, track one, reinforced concrete

Twenty modules, 72 practical subtopics. The full curriculum delivered in the live sessions, module by module.

01 Brief overview of Revit

Difference with AutoCAD software; Benefits over AutoCAD software.

02 Setting up the software

Brief introduction of libraries required; Downloading other extensions from the Autodesk site.

03 Beginning with Revit

Creating a new project; An overview of the templates; An overview of palettes, tabs and panels.

04 Creating datum elements

Creating levels; Creating grids; Propagating grid extents; Customizing grid bubble shape and appearance; Linking CAD files to create grids.

05 Overview of creating 3D elements

Foundations, columns, beams and slabs.

06 Creating columns

Placement of columns; Loading column families.

07 Creating foundations

Hosting foundations on the columns created; Creating strip footing; Using the eccentricity option; Basement floor slab; Loading foundation families into your drawings.

08 Creating beams and slabs

Copying beams and slabs from level to level; Creating a slab; Hollow pot and waffle slabs.

09 Curved beams

Procedure for modeling curved beams.

10 Beam system layout and justification rules

Using the beam system to model structural beams; Modeling beams using beam system layout rules; Modeling beams using beam system justification rules; Deselecting the beam system.

11 Creating staircases and ramps

Types of staircases; Creating an architectural ramp and a structural ramp; Modeling a staircase as a ramp; Shaft openings.

12 Visibility/graphics and filters

Applying visibility/graphic overrides , method 1; Applying visibility/graphic overrides , method 2; Applying visibility/graphic overrides , method 3; Applying visibility/graphic overrides , method 4.

13 Creating roofs and truss placement

Sloped and gable ended roofing; Adding system trusses to your roof; Editing and creating customized trusses.

14 Placing reinforcement

Creating sections in your drawing; Foundation reinforcement; Column reinforcement; Beam reinforcement; Slab reinforcement; Staircase reinforcement; Ramp reinforcement.

15 Annotations

Adding dimensions; Adding spot elevations; Adding spot coordinates; Adding spot slope.

16 Detailing of sections

Placing breaklines in sections; Using hatch files to create regions; Placement of rebar tags on reinforcement; Adding comments on reinforcement; Tagging structural elements.

17 Schedules and material takeoffs

Creating bar bending schedules; Loading shape images into bar bending schedules; Creating material takeoffs for structural elements; Sorting and filtering fields in schedules.

18 Placing drawings on sheets

Loading different paper sizes for title blocks; Using system loaded title blocks; Editing and customizing title blocks; Adjusting text and fields in a title block; Saving title blocks as a Revit family template; Duplicate options for drawings; Placing drawings on sheets.

19 Printing sheets to PDF format

Setting up sheets before printing; Printing one sheet at a time; Printing several sheets at a time.

20 Converting files into CAD and other formats

Export options for your drawing; Transporting your drawing to AutoCAD; Sending your model for analysis.

04 | OUTCOME

THE RESULT

Result, concrete track

20

MODULES

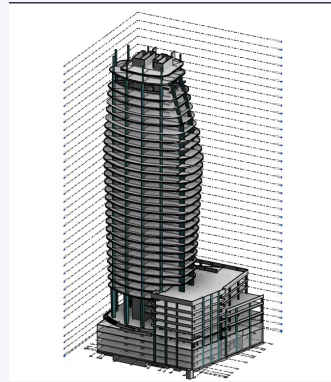
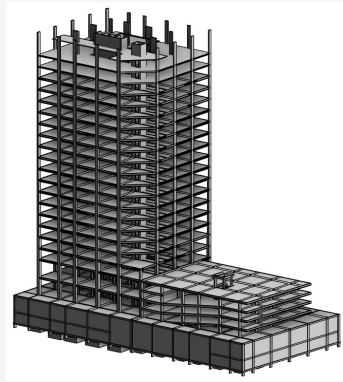
72

SUBTOPICS

4 wks

DURATION

Learners complete the track able to model reinforced concrete frames from datum to documentation, with reinforcement, bar bending schedules and printed sheets. The structures below are modeled during the program.



03 | TRACK 02

THE SOLUTION

The solution, track two, steel structures

Twenty nine modules, 101 practical subtopics. The full curriculum delivered in the live sessions, module by module.

01 Brief overview of Revit

Difference with CAD software; Benefits over CAD software.

02 Setting up the software

Brief introduction of libraries required; Downloading other extensions from the Autodesk site.

03 Beginning with Revit

Creating a new project; An overview of the templates; An overview of palettes, tabs and panels.

04 Creating datum elements

Creating levels; Creating grids; Elevation markers; Propagating grid extents; Customizing grid bubble shape and appearance (optional); Linking CAD files to create grids.

05 Modeling steel columns

Placement of columns; Loading column families.

06 Modeling concrete columns

Placement of concrete columns; Loading column families.

07 Modeling the foundations

Hosting foundations on the columns created; Loading other foundation families into your drawings.

08 Modeling the rafters

Placement of rafters; Loading steel beam sections; Attaching columns to the rafters; Copying rafters from grid to grid.

09 Modeling the steel beams

Loading steel beam sections; Modeling the floor beams; Justification rule of the beams; Copying the floor beams; Modeling beams at the eaves level.

10 Beam system layout and justification rules

Using the beam system to model structural beams; Modeling beams using beam system layout rules; Modeling beams using beam system justification rules; Deselecting the beam system.

11 Composite slab and composite beam

Modeling the composite beam; Modeling the composite slab; Cantilevering the composite slab.

12 Floor openings

Using the shaft opening tool.

13 Curved beams

Procedure for modeling curved beams.

14 Sloped / slanted beams

Modeling sloped beams using the offset tool; Modeling sloped beams using the framing elevation.

15 Purlins and side rails

Modeling purlins using reference planes; Modeling purlins using the beam system; Modeling the side rails.

16 Bracings

Vertical bracings; Horizontal bracing.

17 Visibility/graphics and filters

Applying visibility/graphic overrides , method 1; Applying visibility/graphic overrides , method 2; Applying visibility/graphic overrides , method 3; Applying visibility/graphic overrides , method 4; Hiding elements.

18 Creating sections

Types of sections; Creating sections in your drawing.

19 Placing reinforcement

Setting up the reinforcement cover; Placing reinforcement at the foundation; Column reinforcement; Slab reinforcement.

20 Splitting elements

Splitting elements.

21 Introduction to connections

Uploading the connections; Applying connections to structural elements; Propagating the connections.

22 Connection types

Main types of connection; Haunch connection; Apex connection; Base plate connection; Gusset plate bracing connection; Gusset plate column base plate connection; Purlin / side rail connection; Single sided end plate (simple connection); Single sided end plate (moment connection); Double sided end plate (simple connection); Fin plate (simple connection); Clip angle (simple connection); Column splice connection; Creating custom connections.

23 Fabrication elements, modifiers and parametric cuts

Use and location.

24 Annotations

Adding dimensions; Adding spot elevations; Adding spot coordinates; Adding spot slope; Creating custom bolt and plate tags.

25 Detailing of sections

Placing breaklines in sections; Using hatch files to create regions; Placement of rebar tags on reinforcement; Adding comments on reinforcement; Tagging structural elements.

26 Schedules and material takeoffs

Creating bar bending schedules; Loading shape images into bar bending schedules; Creating material takeoffs for structural elements; Sorting and filtering fields in schedules.

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Loading different paper sizes for title blocks; Using system loaded title blocks; Editing and customizing title blocks; Adjusting text and fields in a title block; Saving title blocks as a Revit family template; Duplicate options for drawings; Placing drawings on sheets.

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Export options for your drawing; Transporting your drawing to AutoCAD; Sending your model for analysis.

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MODULES

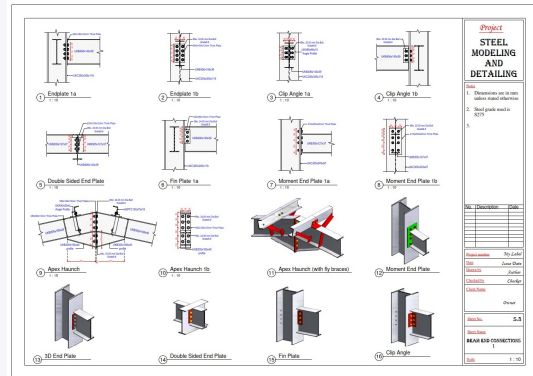
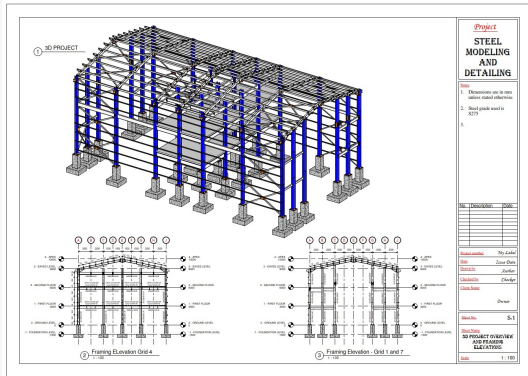
101

SUBTOPICS

4 wks

DURATION

Learners complete the track able to model complete steel structures: rafters, floor beams, purlins and bracing, detailed down to haunch, apex, base plate and gusset plate connections, ready for fabrication data. The structures shown above are modeled during the program.



STEEL AND CONCRETE STRUCTURAL MODELING, AUTODESK REVIT

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ENROL, NEXT COHORT 14 SEPT 2026

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